

Flood Probability Prediction using Data Mining

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Course: CSE 572 - Data Mining by Yeonjung Lee

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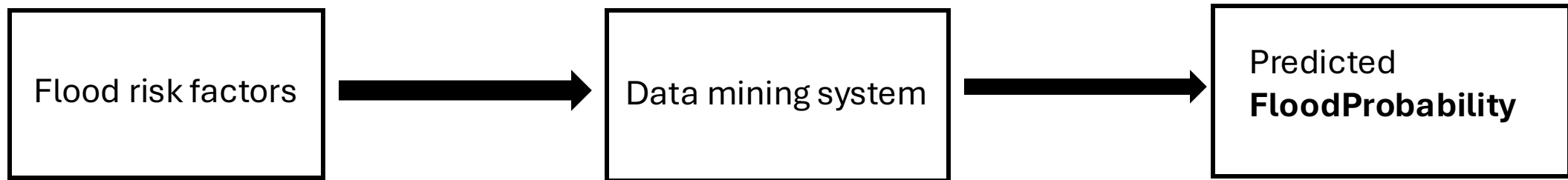
Background

- Floods are one of the most devastating natural disasters, impacting lives of many, destroying infrastructure, and economy of a region.
- Flooding risks have been rising around the world, driven by rapid urbanization, climate change and mismanagement of natural resources.
- This project uses a total of **1.1 million records with 22 predictors** to investigate environmental and human related drivers on **FloodProbability**.



Problem Statement

- The goal of this project is to:
 - Study how multiple factors contribute to **FloodProbability**.
 - **Identify critical drivers** among 21 human + environmental features.
 - **Build predictive models** to forecast flood risk.



Importance of Addressing the Problem

- Floods cause huge economic loss & threaten lives.
- Early prediction enables better disaster preparedness.
- Helps policymakers prioritize interventions (urban planning, drainage, dam safety).

Data Exploration (Initial Results)

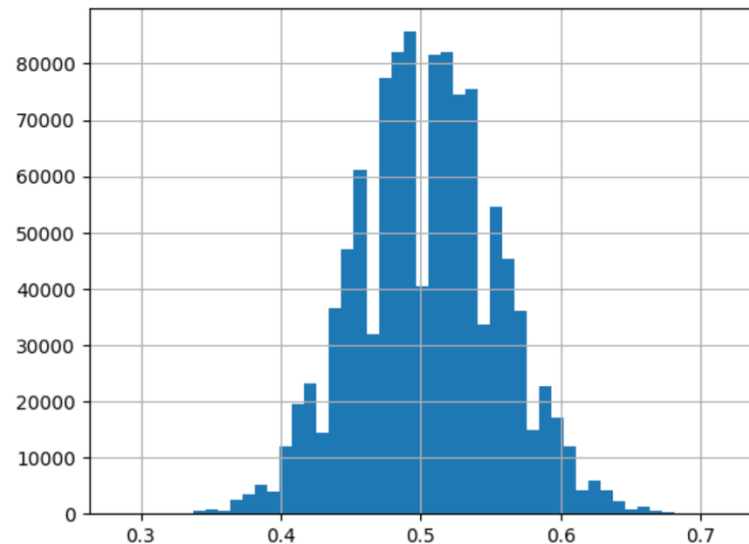
- **Dataset:** 1,117,957 rows, 22 columns, all numeric, no missing values.
- Target (**FloodProbability**): ~normal distribution, mean 0.5.
- Outliers exist but may represent **true extreme cases**.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1117957 entries, 0 to 1117956
Data columns (total 22 columns):
#   Column                                  Non-Null Count  Dtype
---  -
0   id                                     1117957 non-null  int64
1   MonsoonIntensity                     1117957 non-null  int64
2   TopographyDrainage                   1117957 non-null  int64
3   RiverManagement                     1117957 non-null  int64
4   Deforestation                       1117957 non-null  int64
5   Urbanization                        1117957 non-null  int64
6   ClimateChange                       1117957 non-null  int64
7   DamsQuality                         1117957 non-null  int64
8   Siltation                           1117957 non-null  int64
9   AgriculturalPractices                1117957 non-null  int64
10  Encroachments                       1117957 non-null  int64
11  IneffectiveDisasterPreparedness      1117957 non-null  int64
12  DrainageSystems                      1117957 non-null  int64
13  CoastalVulnerability                 1117957 non-null  int64
14  Landslides                          1117957 non-null  int64
15  Watersheds                          1117957 non-null  int64
16  DeterioratingInfrastructure          1117957 non-null  int64
17  PopulationScore                     1117957 non-null  int64
18  WetlandLoss                         1117957 non-null  int64
19  InadequatePlanning                  1117957 non-null  int64
20  PoliticalFactors                     1117957 non-null  int64
21  FloodProbability                     1117957 non-null  float64
```

```
[22] df['FloodProbability'].hist(bins=50)
```

<Axes: >



```
df.isnull().sum()
```

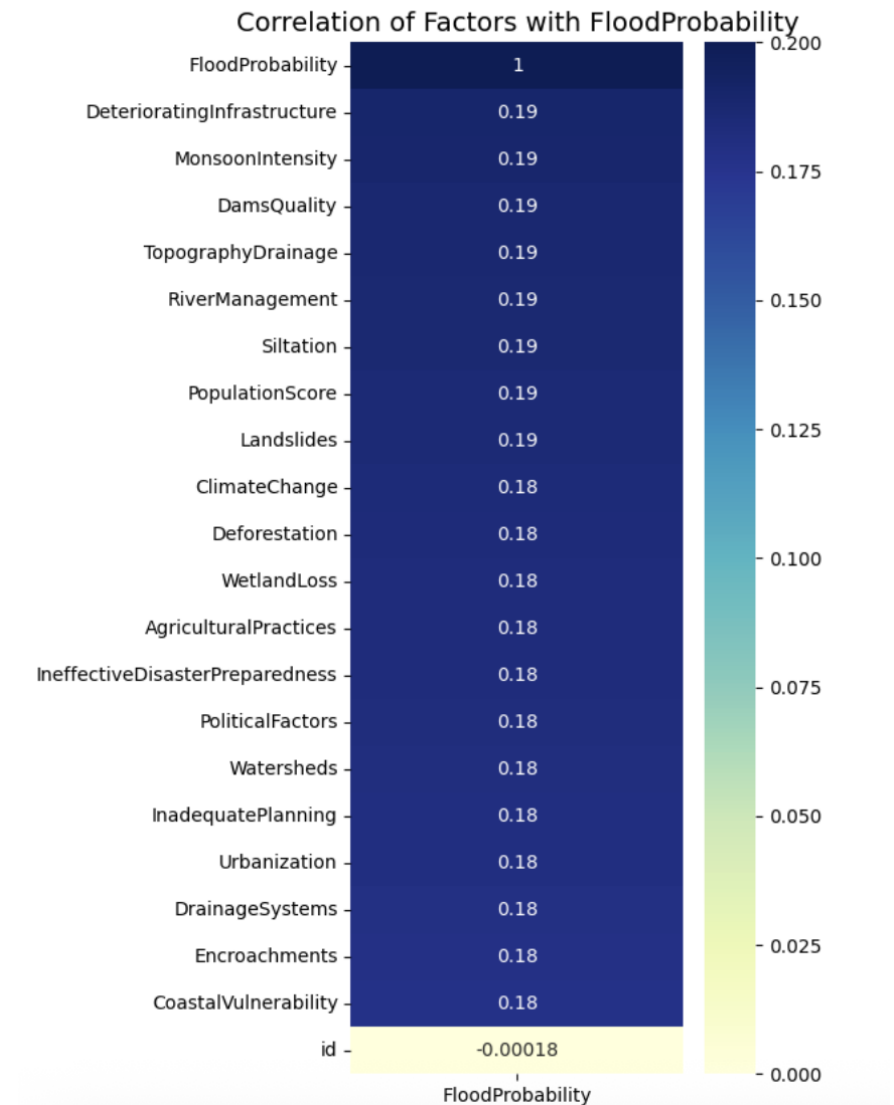
	0
id	0
MonsoonIntensity	0
TopographyDrainage	0
RiverManagement	0
Deforestation	0
Urbanization	0
ClimateChange	0
DamsQuality	0
Siltation	0
AgriculturalPractices	0
Encroachments	0
IneffectiveDisasterPreparedness	0
DrainageSystems	0
CoastalVulnerability	0
Landslides	0
Watersheds	0
DeterioratingInfrastructure	0
PopulationScore	0
WetlandLoss	0
InadequatePlanning	0
PoliticalFactors	0
FloodProbability	0

dtype: int64

Challenges

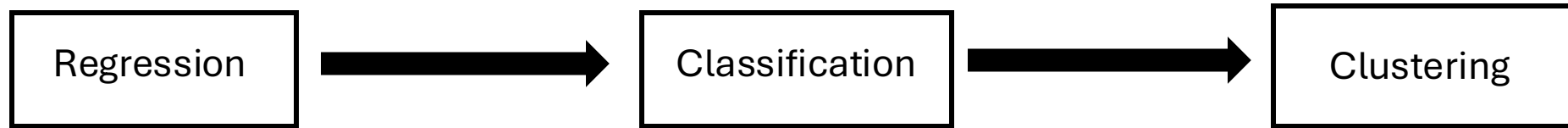
- **Data Challenges:**
 - **Scale:** 1.1M+ rows → computational cost.
 - All correlations with FloodProbability are weak (<0.20), meaning prediction requires **non-linear models**.
- **Modeling Challenges:**
 - Need to handle **multi-factor interactions, interpretability**.
 - Interpretability is critical for policy recommendations.

```
plt.figure(figsize=(4,10))
sns.heatmap(corr, annot=True, cmap="YlGnBu", cbar=True, vmin=0, vmax=0.2)
plt.title("Correlation of Factors with FloodProbability", fontsize=14)
plt.show()
```



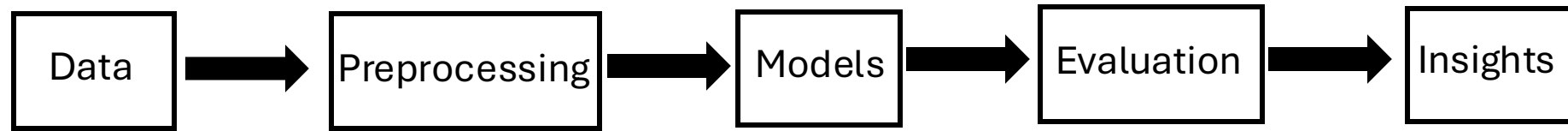
Essential Tasks

- **Primary:** Regression (predict continuous FloodProbability).
- **Secondary:** Classification (low/medium/high risk).
- **Extension:** Clustering regions with similar risk profiles.



Data mining system pipeline

- Data exploration & preprocessing.
- Model building (Logistic Regression, Random Forest, XGBoost).
- Model evaluation & comparison.
 - Evaluate using **R^2** , **RMSE** (for regression) and **F1**, **ROC-AUC** (if classification)
- Translate results into policy insights.



What are your data preprocessing steps?

- Remove irrelevant ID field.
- **Normalization/Standardization** of features for fair comparison.
- **Correlation analysis** to drop redundant predictors.
- **Feature selection** (e.g., combining related features such as “Drainage Systems + Urbanization”).
- **Outlier handling** (e.g., extreme monsoon intensity values).
- **Train-test split** for validation.

After preprocessing, are your data ready for modeling?

- Yes — dataset is clean, balanced, and structured.
- Suitable for regression/classification models.
- Next step: compare Logistic Regression, Random Forest, XGBoost.

Link to video presentation of project pitch: https://drive.google.com/file/d/1a_Mmv9b9X9q-LEaaG16eeVSUStLooZ7Z/view?usp=drive_link

THANK YOU